frontiers

CivicWorkOS: Capability-Preserving Adaptive Task Allocation Among Humans, AI Agents, and Robots in Municipal Services

This Supplementary Material carries the full evaluation protocol and its extended results (Section S1), the proofs and notation (Section S2), framework specification detail (Section S3), the distributional constraints and their legal and data-protection analysis (Section S4), further arithmetic of the worked allocation (Section S5), the elicitation protocol for the judgment parameters (Section S6), and an extended survey of related work (Section S7). Section and equation numbers without the prefix S refer to the main article.

S1 THE EVALUATION PROTOCOL IN FULL

This section records every element of the protocol that Section 5 of the main article summarizes, so that it can be rerun without reference to the code, and then reports the extended results of its reference execution.

S1.1 Testbed Dynamics

Time advances in working days, 250 to a year, for 750 days. Each day, in order: machine capacity is reset, and its reliability belief is propagated and, if a telemetry reading is available, updated; practitioners' daily hours are reset and their offset variances propagated; the day's arrivals join the queue; the queue is sorted by deadline, then by descending task risk, then by arrival; each queued task is offered to the policy, which returns a staffed mode or holds the task; a held task stays queued until its deadline. At the end of the day each active practitioner's true workload index is recorded, a physiological workload reading is taken where available, the recency clock e_w is advanced or reset, a developing practitioner who has reached h_k is certified and a new trainee hired into the post, and a certified practitioner whose exit time has passed leaves.

Each practitioner draws, once per seed, a learning-rate multiplier $\exp(0.25\,\epsilon - 0.03125)$, a retention time constant $120\exp(0.30\,\epsilon - 0.045)$ working days, a workload response $\max(0.6,\ 1 + 0.15\,\epsilon)$ with $\epsilon \sim \mathcal{N}(0,1)$ drawn independently for each, and an exit time after certification that is exponential with mean $250/r_k$ days. Initially certified practitioners start with credited practice uniform on $[1.6, 2.6]\,h_k$ and a recency gap uniform on $\{0,\ldots,9\}$ days; trainees start uniform on $[0, 0.5]\,h_k$. Hires draw from a per-domain pool in a fixed order, so the n-th hire is the same person under every policy. Machine reliability follows $x_{t+1} = \bar{x} + 0.95(x_t - \bar{x}) + 0.004\,\epsilon_t$ around the nominal value of Table 3 and is set to the disruption value inside the disruption window.

A task attempt in staffed mode (m, a) consumes d_i times the mode's duration multiplier in clock hours from each roster member and, where the mode has a machine component, from the domain's machine capacity. With probability equal to the fault rate a machine component faults; otherwise the outcome is Bernoulli with probability $\sigma(\beta_0 z)$ of Equation (13) at the true latent values. A failed attempt causes an incident with probability $0.15\,risk_i$ and the task is re-queued. A fault in a collaborative mode is taken over by the lead, with no reach time and half the situation-awareness time; a fault in a machine-only mode triggers the policy's takeover rule. A takeover succeeds if its true takeover time is within T_i^{safe} and a Bernoulli draw with probability $\sigma(\beta_0(\vartheta_w - b_i))$, reduced by 30% if the operator is overloaded, succeeds. A failed takeover causes an incident with probability $0.50\,risk_i$; a minimal-risk stop causes one with probability $0.03\,risk_i$. Each task may be attempted at most eight times. Table S1 lists the remaining parameters.

Table S1. Global parameters of the reference testbed. All are illustrative modeling choices. Domain parameters are in Table 3 of the main article.

Symbol	Meaning	Value
—	Horizon; working days per year	750; 250
$\bar{d}$	Daily hours each practitioner gives the task stream; overtime cap	4.5 h; $1.3\,\bar{d}$
β_0	Slope of the logistic outcome model	8.0
ϑ_0; ϑ^{comp}; ϑ^{res}	Novice, competence, and residual proficiency	0.30; 0.70; 0.45
τ^{ret}	Nominal retention time constant	120 working days
φ^{led}	Human-led threshold on $\phi_m \omega_{a,w}$	0.30
ω; κ^{sup}	Second member's work share; supervision recovery	0.50; 0.75
L^{max}; ϱ^{L}; ϱ^{to}	Workload ceiling; planning margin; takeover headroom	1.00; 0.95; 1.10
ϖ_m	Human demand per clock hour: H; inspection H; $H{+}R$, $H{+}A$; $H{+}A{+}R$	1.00; 0.85; 0.70; 0.60
—	Duration multiplier: inspection H; $H{+}A{+}R$; others	1.20; 0.80; 1.00
—	Known reliability bonus of $H{+}A{+}R$ over $H{+}R$	0.05
—	Fault rate, routine; in disruption	0.02; 0.25
T^{det}; T^{reach}; T^{sa}	Detection, reach, and situation-awareness time	2; 3; 8 min
χ^{ϑ}, ϑ^{hi}; χ^{ootl}, e^{ootl}; χ^{L}, L^{hi}	Takeover-time factors in Equation (25)	3, 0.9; 0.6, 5 days; 1.5, 0.9
—	Completion value; wage certified; wage trainee; incident cost	10; 1.0/h; 0.6/h; 30
s_0^2	Prior variance: proficiency offset; machine reliability; requirement	0.0025 each
ς^2	Daily process variance: proficiency offset; machine reliability	10^{-4}; 2×10^{-4}
—	Skill-probe probability per human-involved task; noise s.d.	0.30; 0.08
—	Telemetry probability per day; noise s.d.	0.50; 0.10
—	Physiological workload reading probability per worked day; noise s.d.	0.70; 0.10
η_k	Replacement factor of the capability budget	1.2
q^{Z3}; u^{max}	Oversight gate: consequence (consequential flag); predictive s.d.	flag; 0.06
z	Readiness and safety-floor confidence multipliers	1.28; 1.645
g^{ret}	Share of the estimated recency interval at which a practitioner is due	0.50
S_i^{min}	Safety floor on predicted success	$0.55 + 0.35\,risk_i\,crit_i$

S1.2 Policies in Detail

All policies share the queue order, the hours cap, machine capacity, the policy twin (no machine-only inspection), and the safety floor S_i^{min}. For each staffed mode the lead is the certified practitioner with the highest estimated proficiency who fits (Static: the least-loaded), and the trainee is the developing practitioner with fewest hours that day. The expected value of a staffed mode is $10\,(0.5 + 0.5\,crit_i)\,\hat{p} -$ cost $- 30 \times 0.15\,risk_i(1 - \hat{p})$.

Static selects the fixed mode of its domain, preferring the roster with a trainee when one is free, and falls back to any feasible staffed mode when the fixed one is infeasible. It plans on the nominal proficiency model and the commissioning ratings and never updates. Its takeover operator is the least-loaded certified practitioner. *Efficiency-focused* selects the feasible staffed mode of highest expected value, checks the safety floor on the posterior mean, updates beliefs from every outcome and signal, and hands takeovers to the certified practitioner of highest estimated proficiency. *CivicWorkOS* additionally: checks the safety floor on the lower confidence bound; applies the oversight gate of Equation (23); plans to the workload ceiling of Equation (21) on learned workload responses; adds rosters led by the most-due current

practitioner and supervised refreshers for lapsed ones; selects through the retention and budget stages of Algorithm 1; and transfers control only within the readiness set of Equation (24), otherwise commanding a minimal-risk stop. The two ablations remove, respectively, all belief updating (planning on the nominal model and commissioning ratings, with workload responses fixed at one) and the retention stage with its due and refresher rosters. The exploratory outcome-only variant of the main article keeps every CivicWorkOS mechanism and the outcome update of Equation (14), but removes all three multimodal channels, so its workload responses stay at their prior mean of one.

S1.3 Metric Definitions

Completion rate is completed tasks over arrived tasks; a task counts as completed when a success occurs on or before its deadline day. The window variant restricts both counts to tasks arriving in the event window. *Cost per completed task* is total cost (wages for every human clock hour assigned, including takeovers, machine hours, and incident losses) over completed tasks; the *cost index* divides it by the efficiency-focused value on the same seed and scenario and multiplies by 100. *Overload share* is the share of active practitioner-days whose true workload index exceeds L^{max}. *Current share* is the share of active certified practitioners whose true proficiency is at least ϑ^{comp} on the last day; *certified* and *current* counts are reported with it. *Takeover success* is the share of machine faults recovered by a human within the safety time budget; a minimal-risk stop counts as not recovered. The window variant counts faults occurring in the window. *Incidents* are per 1,000 arrived tasks. Two read-only accuracy metrics, added during review, never enter a decision: the *proficiency estimate MAE* is the mean absolute difference between the allocator's estimate $\hat{\vartheta}_w$ and the true proficiency over active practitioners on the last day, and the *reliability estimate MAE* is the mean absolute difference between the belief mean and the true reliability of each domain's fleet or agent pool, averaged over days, overall and in the event window.

S1.4 Pre-Specified Hypotheses and Deviations

The hypotheses of Table 8 were written to `configs/protocol/hypotheses.yaml` before the evaluation code existed. During development on seeds 1000–1009 four implementation defects were found and corrected before any reporting run: (i) the retention rule ranked leads by recency need without excluding practitioners already estimated to have lapsed, which selected unqualified leads; (ii) the recency clock added the learned offset to the attainable proficiency, double-counting decay; (iii) the inspection domain had no human-only fallback mode with feasible duration, so a detected fleet failure left no mode to which control could be transferred; and (iv) the retention stage was filtered from the budget catch-up subset, which excluded refresher rosters whenever a domain was behind its trainee trajectory and allowed estimated lapses to become self-perpetuating. Each correction made the implementation match the mechanism specified in the main article, and none changed a hypothesis, threshold, metric, or seed; the file's `DEVIATIONS` list is empty. The development process was not blind, however: the defects were found by inspecting development-seed outputs that included comparisons between policies, so the reader should treat the implementation as having been debugged with those comparisons in view. The guard against this is that the reporting seeds were run once, after the configuration was frozen, and that every pre-specified hypothesis is reported whatever its outcome. During review, the reporting runs were executed a second time to add three read-only accuracy metrics (proficiency and reliability estimation errors), which never enter a decision; every value reported before that addition, 11,700 in all, was identical across the two executions. The exploratory analyses of the main article reuse the reporting seeds and are labeled exploratory wherever they appear.

S1.5 Extended Results

Tables S2–S4 give every metric of the reference execution with its 95% t-interval over 30 seeds. Window metrics are omitted from the routine table, which has no event window. All entries are generated from `Paper/Frontiers/data/summary.json` by `make_sim_tables.py`.

Table S2. Reference execution, routine operations: means with 95% intervals over 30 seeds.

Metric	Static	Efficiency	CivicWorkOS	No learning	No retention
Task completion (%)	99.4 [99.3, 99.5]	99.3 [99.2, 99.4]	97.8 [96.8, 98.8]	99.5 [99.5, 99.6]	96.2 [95.0, 97.4]
Service cost index	123.6 [122.7, 124.5]	100.0 [100.0, 100.0]	173.4 [166.4, 180.4]	180.2 [178.5, 181.9]	159.0 [152.4, 165.5]
Overloaded days (%)	0.11 [0.05, 0.17]	4.18 [3.33, 5.03]	0.04 [0.03, 0.05]	6.49 [5.64, 7.33]	0.04 [0.03, 0.05]
Mean workload index	0.141 [0.138, 0.143]	0.146 [0.142, 0.150]	0.406 [0.391, 0.422]	0.428 [0.424, 0.432]	0.362 [0.347, 0.378]
Certified, end	25.7 [24.6, 26.7]	23.0 [22.1, 23.9]	32.8 [31.3, 34.4]	33.9 [32.4, 35.4]	32.9 [31.4, 34.4]
Current, end	14.2 [13.6, 14.8]	9.6 [9.1, 10.0]	27.5 [25.9, 29.1]	28.4 [27.2, 29.6]	14.1 [13.1, 15.1]
Current share, end	0.558 [0.530, 0.586]	0.418 [0.398, 0.438]	0.838 [0.805, 0.871]	0.841 [0.822, 0.859]	0.427 [0.404, 0.450]
Mean proficiency of certified, end	0.704 [0.695, 0.714]	0.667 [0.656, 0.677]	0.839 [0.827, 0.851]	0.849 [0.844, 0.854]	0.679 [0.667, 0.692]
Certifications	2.8 [2.4, 3.2]	0.0 [0.0, 0.0]	10.9 [10.1, 11.7]	12.3 [11.6, 13.0]	10.9 [10.1, 11.7]
Trainee credit (h/yr)	760 [758, 763]	0 [0, 0]	1961 [1855, 2068]	2161 [2100, 2222]	1955 [1845, 2065]
Takeover success, all	0.876 [0.869, 0.883]	0.918 [0.912, 0.924]	0.893 [0.880, 0.906]	0.897 [0.885, 0.908]	0.790 [0.748, 0.832]
Takeover time (min)	14.13 [13.98, 14.28]	13.31 [13.20, 13.42]	12.98 [12.80, 13.16]	12.71 [12.62, 12.79]	13.04 [12.87, 13.22]
Minimal-risk stops	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	26.8 [17.9, 35.7]	9.8 [2.3, 17.2]	106.8 [75.5, 138.1]
Incidents per 1,000	7.83 [7.70, 7.96]	6.32 [6.21, 6.43]	5.57 [5.41, 5.73]	6.22 [6.10, 6.33]	5.45 [5.29, 5.62]
Proficiency estimate MAE, end	0.023 [0.021, 0.024]	0.020 [0.019, 0.022]	0.023 [0.022, 0.024]	0.035 [0.033, 0.037]	0.028 [0.026, 0.030]
Reliability estimate MAE	0.010 [0.010, 0.010]	0.053 [0.052, 0.055]	0.030 [0.029, 0.031]	0.010 [0.010, 0.010]	0.030 [0.029, 0.031]

Table S3. Reference execution, demand surge: means with 95% intervals over 30 seeds. Window metrics refer to the 30-day surge.

Metric	Static	Efficiency	CivicWorkOS	No learning	No retention
Task completion (%)	98.2 [98.1, 98.4]	98.2 [97.9, 98.5]	97.7 [97.1, 98.4]	99.1 [99.0, 99.3]	96.0 [95.2, 96.8]
Window completion (%)	88.2 [87.6, 88.8]	87.7 [87.0, 88.3]	92.7 [91.9, 93.4]	96.6 [96.0, 97.1]	85.6 [84.4, 86.8]
Service cost index	123.8 [122.2, 125.3]	100.0 [100.0, 100.0]	176.6 [172.6, 180.7]	181.3 [178.6, 184.1]	162.2 [157.7, 166.7]
Overloaded days (%)	0.11 [0.04, 0.18]	3.09 [2.35, 3.83]	0.04 [0.03, 0.04]	5.60 [4.77, 6.43]	0.03 [0.03, 0.04]
Mean workload index	0.144 [0.141, 0.146]	0.144 [0.139, 0.149]	0.417 [0.407, 0.426]	0.435 [0.430, 0.439]	0.372 [0.362, 0.383]
Certified, end	25.0 [23.6, 26.3]	22.1 [20.8, 23.3]	31.7 [29.9, 33.6]	33.1 [31.2, 35.0]	31.7 [29.8, 33.6]
Current, end	13.4 [12.4, 14.4]	8.7 [8.1, 9.3]	27.7 [25.9, 29.5]	28.4 [26.8, 30.0]	14.3 [13.3, 15.3]
Current share, end	0.538 [0.506, 0.570]	0.395 [0.378, 0.412]	0.872 [0.852, 0.891]	0.858 [0.841, 0.874]	0.454 [0.428, 0.480]
Mean proficiency of certified, end	0.690 [0.678, 0.702]	0.655 [0.646, 0.664]	0.851 [0.842, 0.859]	0.851 [0.845, 0.857]	0.687 [0.673, 0.702]
Certifications	3.0 [2.7, 3.4]	0.0 [0.0, 0.0]	10.6 [9.9, 11.3]	11.9 [11.1, 12.6]	10.5 [9.8, 11.3]
Trainee credit (h/yr)	794 [790, 798]	0 [0, 0]	2005 [1923, 2087]	2142 [2068, 2215]	1991 [1904, 2078]
Takeover success, all	0.879 [0.872, 0.886]	0.920 [0.917, 0.924]	0.909 [0.901, 0.918]	0.897 [0.889, 0.905]	0.786 [0.755, 0.817]
Takeover success, window	0.879 [0.867, 0.892]	0.891 [0.877, 0.906]	0.827 [0.791, 0.863]	0.781 [0.723, 0.840]	0.414 [0.338, 0.490]
Takeover time (min)	14.19 [14.03, 14.35]	13.34 [13.27, 13.42]	12.97 [12.80, 13.15]	12.73 [12.65, 12.82]	12.97 [12.74, 13.19]
Minimal-risk stops	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	18.4 [13.8, 22.9]	13.8 [8.9, 18.6]	115.0 [92.3, 137.8]
Incidents per 1,000	7.66 [7.47, 7.85]	6.25 [6.09, 6.41]	5.69 [5.51, 5.87]	6.22 [6.05, 6.39]	5.54 [5.36, 5.72]
Incidents per 1,000, window	5.84 [5.43, 6.26]	4.72 [4.37, 5.06]	5.25 [4.86, 5.64]	5.76 [5.36, 6.15]	4.85 [4.42, 5.28]
Proficiency estimate MAE, end	0.022 [0.021, 0.024]	0.019 [0.017, 0.021]	0.023 [0.022, 0.024]	0.033 [0.031, 0.034]	0.028 [0.026, 0.029]
Reliability estimate MAE	0.010 [0.010, 0.010]	0.053 [0.052, 0.055]	0.030 [0.029, 0.031]	0.010 [0.010, 0.010]	0.031 [0.030, 0.031]
Reliability estimate MAE, window	0.010 [0.009, 0.011]	0.057 [0.054, 0.060]	0.032 [0.029, 0.035]	0.010 [0.009, 0.011]	0.033 [0.030, 0.035]

Table S4. Reference execution, automation disruption: means with 95% intervals over 30 seeds. Window metrics refer to the 20-day disruption.

Metric	Static	Efficiency	CivicWorkOS	No learning	No retention
Task completion (%)	98.7 [98.5, 98.8]	97.1 [97.0, 97.3]	96.6 [95.6, 97.6]	99.4 [99.3, 99.4]	94.2 [93.0, 95.3]
Window completion (%)	76.4 [75.4, 77.5]	26.8 [25.6, 28.0]	71.9 [68.0, 75.9]	93.5 [91.8, 95.1]	52.4 [48.4, 56.3]
Service cost index	125.7 [124.4, 127.0]	100.0 [100.0, 100.0]	171.6 [165.1, 178.2]	180.2 [178.6, 181.8]	156.3 [150.2, 162.5]
Overloaded days (%)	0.42 [0.34, 0.51]	4.39 [3.53, 5.25]	0.04 [0.04, 0.05]	6.59 [5.75, 7.44]	0.04 [0.03, 0.05]
Mean workload index	0.150 [0.147, 0.153]	0.150 [0.146, 0.154]	0.410 [0.395, 0.425]	0.432 [0.428, 0.436]	0.360 [0.345, 0.376]
Certified, end	25.7 [24.6, 26.7]	23.0 [22.1, 23.9]	32.9 [31.4, 34.4]	33.9 [32.4, 35.4]	32.8 [31.3, 34.3]
Current, end	14.4 [13.6, 15.2]	9.6 [9.1, 10.1]	27.6 [25.9, 29.2]	28.5 [27.3, 29.7]	13.4 [12.4, 14.4]
Current share, end	0.565 [0.529, 0.601]	0.418 [0.398, 0.438]	0.839 [0.805, 0.873]	0.844 [0.825, 0.862]	0.407 [0.382, 0.432]
Mean proficiency of certified, end	0.714 [0.700, 0.728]	0.667 [0.657, 0.677]	0.839 [0.827, 0.851]	0.850 [0.845, 0.855]	0.671 [0.658, 0.683]
Certifications	2.8 [2.4, 3.2]	0.0 [0.0, 0.0]	11.0 [10.2, 11.8]	12.3 [11.6, 13.0]	10.8 [10.0, 11.6]
Trainee credit (h/yr)	799 [792, 806]	0 [0, 0]	1962 [1860, 2065]	2161 [2100, 2222]	1937 [1829, 2045]
Takeover success, all	0.876 [0.869, 0.882]	0.912 [0.906, 0.918]	0.891 [0.879, 0.903]	0.892 [0.874, 0.910]	0.790 [0.749, 0.831]
Takeover success, window	0.874 [0.866, 0.882]	0.734 [0.699, 0.768]	0.812 [0.774, 0.849]	0.878 [0.839, 0.917]	0.720 [0.658, 0.781]
Takeover time (min)	14.12 [13.98, 14.25]	13.42 [13.32, 13.52]	12.78 [12.61, 12.94]	12.74 [12.64, 12.84]	12.87 [12.70, 13.04]
Minimal-risk stops	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	29.4 [20.7, 38.1]	21.9 [4.4, 39.4]	107.9 [76.9, 138.9]
Incidents per 1,000	8.57 [8.43, 8.71]	6.22 [6.12, 6.33]	5.50 [5.35, 5.66]	6.90 [6.77, 7.03]	5.34 [5.17, 5.50]
Incidents per 1,000, window	35.76 [33.88, 37.64]	3.76 [3.15, 4.36]	5.02 [4.31, 5.73]	30.96 [29.05, 32.86]	3.60 [2.98, 4.22]
Proficiency estimate MAE, end	0.026 [0.024, 0.027]	0.020 [0.018, 0.022]	0.022 [0.021, 0.023]	0.035 [0.033, 0.036]	0.028 [0.026, 0.030]
Reliability estimate MAE	0.021 [0.020, 0.021]	0.058 [0.056, 0.059]	0.036 [0.035, 0.037]	0.021 [0.020, 0.021]	0.036 [0.036, 0.037]
Reliability estimate MAE, window	0.403 [0.403, 0.403]	0.131 [0.127, 0.134]	0.145 [0.141, 0.149]	0.403 [0.403, 0.403]	0.147 [0.142, 0.151]

S2 PROOFS, DUALS, AND NOTATION

S2.1 Proofs of the Propositions

Proof of Proposition 1.

First take ϑ_w^{∞} fixed during a gap. Equation (7) is then strictly decreasing in e_w, and setting $\vartheta_w = \vartheta^{comp}$ and solving gives $e_w = \bar{e}_w$ of Equation (10), so $\vartheta_w \geq \vartheta^{comp}$ exactly when $e_w \leq \bar{e}_w$. A human-led task resets e_w to zero, so the practitioner is current throughout if and only if no gap exceeds $\bar{e}_w$. Over a period ΔT, at least $\lceil \Delta T / \bar{e}_w \rceil$ resets are needed per practitioner; summing over N_k practitioners, each with interval at least $\bar{e}$, gives the volume bound. In general ϑ_w^{∞} is non-decreasing during a gap, because tasks below the human-led threshold still credit $\ell_i \phi_m \omega_{a,w} \geq 0$ to c_w. Since $\bar{e}_w$ is increasing in ϑ_w^{∞}, evaluating it at the value at the start of the gap gives an interval that is sufficient for currency; it is also necessary when no such credit accrues, which is the case the proposition states.

Proof of Proposition 2.

Every term on the left-hand side of Equation (19) is bounded by $\ell_i \bar{\zeta}_k$, and Equation (26b) selects one staffed mode per task, so $\Lambda_k \leq \bar{\zeta}_k \sum_i \ell_i = \Phi_k \bar{\zeta}_k$. Substituting the second argument of B_k and solving for r_k yields Equation (29).

Proof of Proposition 3.

A developing practitioner accrues $\ell_i \phi_m \omega_{a,w} = learn_i \, d_i \, \phi_m \, \omega_{a,w}$ credited hours on a task that occupies d_i of their clock hours, so the accrual rate per clock hour is the duration-weighted mean of $learn_i \phi_m \omega_{a,w}$. Under the factorization assumption this is $\overline{learn}_k \bar{\phi}_k \bar{\omega}_k$, giving $\Theta_k \overline{learn}_k \bar{\phi}_k \bar{\omega}_k$ per period. Dividing $h_k = \overline{learn}_k h_k^{raw}$ by this rate cancels $\overline{learn}_k$ and yields Equation (30). The domain completes $B_k / h_k = \eta_k N_k r_k$ practitioners per period, and by Little's law a queue with that throughput and residence time τ_k holds $\eta_k N_k r_k \tau_k$ members.

Proof of Proposition 4.

Immediate from the definition of the infimum and the continuity of Equation (16) in w_9.

Proof of Proposition 5.

The resource test rejects any staffed mode whose consumption exceeds remaining availability, so the running total never exceeds U_r for admitted tasks. For displacement, a staffed mode is admitted only if the realized displacement before the task plus its estimated increment is within τ_g; execution can exceed the estimate by at most $\bar{\delta}_g$, so after any admitted task the displacement is at most $\tau_g + \bar{\delta}_g$. Because each outcome is observed before the next decision, once displacement exceeds τ_g every staffed mode with a non-negative estimated increment is rejected, so the excess cannot grow, independently of the number of arrivals.

The outcome update.

Equation (14) is the standard assumed-density-filtering update for a probit likelihood $\Phi(\kappa y^*)$ with independent Gaussian factors (Minka, 2001): the exact posterior of the linear predictor y^* after observing y is a truncated Gaussian whose first two moments are $\bar{y}^* + (2y - 1)\, v\, \kappa \nu(t) / \sqrt{1 + \kappa^2 v}$ and $v[1 - v\kappa^2 \nu(t)(\nu(t) + t)/(1 + \kappa^2 v)]$, and projecting onto the factors in proportion to $\alpha_q s_q^2$ gives the stated updates. The logistic-to-probit substitution $\sigma(x) \approx \Phi(x\sqrt{\pi/8})$ is the usual one; the predictive success probability used for decisions is $\sigma\big(\beta_0 \bar{y}^* / \sqrt{1 + \pi \beta_0^2 v / 8}\big)$.

S2.2 Duals from a Binary Program

Integer programs have no exact duals. The multipliers in Equation (27) are taken from the LP relaxation at the root node, the standard construction for Lagrangian decomposition of assignment problems, so Equation (27) approximates the relaxed rather than the integral program. The integrality gap the solver reports measures the distance between the two programs' optimal values; it does not bound the error of

the online rule between rebalances. In the worked allocation the gap is 7.8×10^{-6} objective units per task. The reserve term $\Delta_s^{res}(i,m,a) = \frac{d_i}{D_s^{tot}}\left[Res_s^{(n_s)}(t \mid (m,a) \text{ generalized across } s) - Res_s^{(n_s)}(t)\right]$ scales the counterfactual change in surviving capacity, were every task of the service executed in (m,a), by the task's share d_i/D_s^{tot} of the service's task-hours.

S2.3 What a Guarantee for the Online Rule Would Require

Proposition 5 says nothing about how the online rule performs against the program it approximates. The bound to seek is regret of order $O(\sqrt{T})$ in the number of arrivals against the offline optimum, with cumulative violation of the budget and access constraints of the same order. Close relatives have such bounds: a primal–dual rule is $(1 - o(1))$-competitive under random-order arrivals with small requests (Devanur and Hayes, 2009), dual mirror descent attains $O(\sqrt{T})$ regret under stochastic arrivals (Balseiro et al., 2023), online budgeted matching has the adversarial benchmark $1 - 1/e$ (Mehta et al., 2007), and constrained Markov decision processes license the same primal–dual construction (Altman, 1999; Paternain et al., 2019). Four features keep them from applying directly. The duals are refreshed only at rebalances. The capability budget is a *covering* constraint, protected only by reserving future capacity, for which the catch-up test of Equation (28) is a heuristic. Its coefficients are endogenous, because assignments move practitioners across the certification threshold and, with learning, change the beliefs on which the coefficients are evaluated. And the retention constraint (20) is a per-practitioner covering constraint over a sliding window, which couples the online decisions in time. The second, third, and fourth are specific to capability formation and retention, and are where we expect the difficulty to lie; safe reinforcement learning (García and Fernández, 2015) is a further route.

S2.4 Notation

Table S5 collects the symbols. Some letters are shared and distinguished by index: the formation time τ_k, the retention constant τ_w^{ret}, and the transition ceiling τ_g; the debt weight β and the logistic slope β_0; the debt weight ϵ and the access tolerance ε_k; and the planning margins ϱ^L, ϱ^{to}.

Table S5. Notation. Indices: i task, m execution mode, a roster, w practitioner, j AI agent or robot, k capability domain, s critical service, g worker group, r resource, t time.

Symbol	Meaning
$x_{i,m,a}$; $\mathcal{M}$; $\mathcal{A}_{i,m}$	Decision variable, Equation (4); mode set; feasible rosters
T_i; d_i; ℓ_i; b_i, b_k; T_i^{safe}	Task vector and duration, Equation (2); developmental content, Equation (3); requirement level and domain mean; safety time budget
$\omega_{a,w}$; ψ_a; ϕ_m	Work split; developmental eligibility, Equation (9); human developmental share, Equation (5)
c_w; e_w; h_k, h_k^{raw}	Credited practice; working days since the last human-led task; credited and raw hours to certification
ϑ_w^{∞}; ϑ_w; ϑ_0, ϑ^{comp}, ϑ^{res}	Attainable and current proficiency, Equations (6) and (7); novice, competence, and residual levels
τ_w^{ret}; $\bar{e}_w$; φ^{led}; g^{ret}	Retention time constant; recency interval, Equation (10); human-led threshold; due fraction
ϑ_j; o_w; ι_w	Machine reliability; proficiency offset from the nominal model; workload response
μ, s^2; ς^2; σ_n^2	Belief mean and variance; process variance; observation noise of channel n
β_0; $y_{i,m,a}^*$; $\bar{\vartheta}_a$; κ^{sup}	Logistic slope, linear predictor, and team proficiency in Equation (13); supervision recovery
w_{ld}, w_{sp}; $\nu(t)$	Lead and supervised second practitioner of a roster; inverse Mills ratio in Equation (14)
L_w, L^{max}; ϖ_m; $\bar{d}$	Workload index and ceiling; human demand per hour of mode m; available hours
$\mathcal{H}_i(t)$; $\hat{T}_w^{to}(i)$	Readiness set, Equation (24); expected takeover time, Equation (25)
B_k; Λ_k; $\bar{B}_k$; δ_k	Capability budget and accrued practice, Equation (19); pro-rated trajectory; catch-up increment, Equation (28)
η_k, r_k, N_k, N_k^{dev}	Replacement factor, attrition, certified and developing headcount
Φ_k; $\bar{\zeta}_k$; r_k^{crit}; τ_k; $\hat{n}_k$	Developmental supply; largest credited share; critical attrition, Equation (29); time to certification, Equation (30); required cohort, Equation (31)
Θ_k; $\bar{\phi}_k$, $\bar{\omega}_k$	Trainee clock hours per period; mean developmental and work shares (Proposition 3)
F_k; $Res_s^{(n_s)}$, κ_s, D_s^{peak}	Readiness-weighted fallback capacity, Equation (12); surviving capacity, servable share, peak demand, Equation (22)
$Q, S, P, Eq^{srv}, Tr, Cost, En, Pr$	Flow terms of SCV, Equation (16)
D^{skill}, D^{fall}, D^{acct}, D^{dep}, D^{trans}	Components of CAD, Equations (17) and (S1)
$w_1, \ldots, w_9$; $\alpha, \ldots, \epsilon$; $w_9^{\star}$	Objective weights, debt weights, flip threshold (Proposition 4)
λ_k, μ_s, υ_r	Duals of the budget, reserve, and resource constraints
$\Pi_{k,g}$, $\theta_{k,g}$, ε_k; Δ_g, τ_g	Protected-practice share, target, tolerance, Equation (S3); group displacement and ceiling, Equation (S4)

S3 FRAMEWORK SPECIFICATION DETAIL

S3.1 The Debt Components

Equation (18) defines D^{skill}. The other four components of Equation (17) are normalized against externally fixed denominators, Equation (S1), so that a decade of decline registers as a decade of decline rather than being compared with its own recent past:

$$\begin{aligned} D^{fall}_{i,m} &= \min\Big(1,\ \Delta F_{k(i)}(m) \,/\, R^{min}_{\sigma(i)}\Big) && \text{against the statutory reserve,} \\ D^{acct}_{i,m} &= crit_i \cdot n^{unacct}_{i,m} / n^{steps}_{i,m} && \text{against the decision-step count of the task,} \\ D^{dep}_{i,m} &= n^{single}_{i,m} / n^{auto}_{i,m} && \text{against the automated-input count of the mode,} \\ D^{trans}_{i,m,a} &= \textstyle\sum_g \varpi_g(i,m,a)\,\big(1-\chi_g\big) && \text{against funded transition coverage } \chi_g, \end{aligned} \tag{S1}$$

where $\Delta F_k(m)$ is the reduction in fallback capacity mode m would cause if generalized across the domain, n^{unacct} the decision steps lacking a designated accountable human, n^{single} the automated inputs drawn from a single vendor with no qualified alternative (Sculley et al., 2015), and $\varpi_g(i,m,a)$ group g's share of the workers whose hours the staffed mode reduces. The debt framing follows technical debt as the deferred cost of an expedient choice (Cunningham, 1992; Sculley et al., 2015).

S3.2 Populating the Task Vector

Cognitive, physical, and social demand come from standardized task-demand instruments or expert ergonomic rating (El Makrini et al., 2019; Merlo et al., 2023); safety risk from incident history or hazard classification; legal authority and privacy sensitivity from the policy twin's rule encoding; urgency from service-level targets; criticality from continuity planning; and learning value from the elicitation protocol of Section S6. The requirement level b_i and the safety time budget T^{safe}_i are initialized from commissioning ratings and refined by the learning layer.

S3.3 Policy Rules and a Worked Encoding

A policy rule is a tuple $\rho = \langle \text{scope}, \text{guard}, \text{status}, \text{authority}, \text{validity}, \text{provenance} \rangle$. The scope is a conjunction of catalog predicates, the guard a decidable predicate over the task vector, roster, and ledger state, the authority ranks the source from statute to operational guidance, the validity is a time interval, and the provenance names the instrument and the decision that encoded it. Statuses form the total order *allow* $\prec$ *allow-with-oversight* $\prec$ *restrict* $\prec$ *prohibit*, and the status of a staffed mode is the join over the rules that fire, evaluated first at mode level and then at roster level. Because the join ignores authority, no rule can relax another during evaluation; relaxation requires amending or retiring the restrictive rule, on the strength of an instrument of strictly higher authority. *allow-with-oversight* binds a named reviewer to the staffed mode and adds the reviewer's time to its cost; *restrict* narrows the roster set to rosters meeting a minimum human share or stage requirement. A nightly consistency check reports every scope whose join is *prohibit* across all modes.

Article 14 of the EU AI Act (European Parliament and Council of the European Union, 2024) requires effective oversight of high-risk systems by competent natural persons. For a municipal structural-inspection determination the encoding is: scope $\sigma(i) =$ *structural-inspection* $\wedge\ m \in \{A, R, A{+}R, H{+}A, H{+}A{+}R\}$; guard $auth_i \geq 0.8 \wedge crit_i \geq 0.7$; status *restrict* with roster guard $\exists w \in a :$ current$(w) \wedge$ license(w); authority, regulation with its national implementing instrument. A fully autonomous mode is thereby restricted to rosters carrying a licensed current practitioner rather than prohibited outright, the difference between a prohibition and an oversight obligation. Note that the roster guard requires a *current* practitioner, not merely a certified one: with the proficiency model of Section 3.3, the oversight obligation itself depends on recency.

S3.4 Zone Transitions

A task class is demoted one zone toward Z4 on any single condition: an incident of severity 3 or above, or an incident rate above 1.5 times the class baseline over a rolling window of four quarters; an upheld-appeal rate above 0.05; accumulated debt for the class rising for two consecutive quarters; surviving capacity below R_s^{min}; accrued practice below its trajectory for two consecutive quarters; or the learned reliability of the class's automation falling below its commissioning floor with confidence. Promotion requires all of the converse conditions and an affirmative decision of the municipal review panel, because evidence that automation works should not expand it without a political act, while evidence that it fails should contract it at once.

S4 DISTRIBUTIONAL CONSTRAINTS, LAWFULNESS, AND DATA PROTECTION

S4.1 Capability Access and Just Transition

The capability budget decides how much developmental work is protected but not who receives it. If developing practitioners are recruited through the channels that produced the incumbents, a composition-blind budget reproduces the incumbent composition for a full replacement cycle. With group shares $\pi_{a,g}^{dev}$ of a roster's developing members, the protected-practice share of group g is Equation (S2), and the *Capability Access Constraint* is Equation (S3):

$$\Pi_{k,g} = \frac{\sum_{i \in \mathcal{T}_k} \sum_{m,a} x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a\, \pi_{a,g}^{dev}}{\sum_{i \in \mathcal{T}_k} \sum_{m,a} x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a}, \tag{S2}$$

$$\Pi_{k,g} \;\geq\; \theta_{k,g} - \varepsilon_k \qquad \forall\, g \in \mathcal{G}. \tag{S3}$$

The target $\theta_{k,g}$ is not the incumbent composition; it is set from the qualified local labor pool, a statutory equality duty, or an explicit equity target, and published. The constraint is linearized by multiplying through by the domain's practice volume and made elastic with a penalized slack, so an unattainable target yields a published shortfall rather than infeasibility or a forced assignment. Worker groups are a coarsening of the declared protected attributes subject to a minimum cell size n^{min} (30 practitioners per cell per domain by default); smaller cells are reported descriptively and carry no binding constraint.

With $\Delta_g(\mathbf{x})$ the net fractional reduction in the allocated hours of payroll group g relative to its baseline, the *Just Transition Constraint*, following the ILO guidelines (International Labour Organization, 2015), is Equation (S4):

$$\Delta_g(\mathbf{x}) \leq \tau_g \qquad \forall\, g \in \mathcal{G}. \tag{S4}$$

Reskilling coverage is a municipal budget fact rather than a function of $\mathbf{x}$, so it is enforced as a gating predicate in the policy twin: a staffed mode that would raise Δ_g for a group whose funded reskilling coverage is below its floor is *prohibit* until the pathway is funded.

Who receives the protected work in the worked domain. Suppose the 24 inspectors of Section 6.1 are 21 men and three women while the qualified local labor pool is 68% to 32%. No cell of a domain this small meets n^{min}, so a deployment would report shares descriptively; setting that rule aside to show the mechanism, a composition-blind budget sends $0.125 \times 4976.64 = 622$ protected hours a year to the under-represented group, 0.432 certified-equivalents. Setting $\theta_{k,g} = 0.32$ with $\varepsilon_k = 0.05$ requires at least 1,344 hours, 0.933 certified-equivalents, 2.16 times the composition-blind rate, and is met only if the city also recruits 3.8 of the 14 developing inspectors of Equation (37) from that group rather than 1.8 (Table S6). Workers outside the municipal payroll are absent from $\mathcal{W}_k$ and receive a zero share whatever $\theta_{k,g}$ is.

S4.2 Lawfulness of the Access Constraint

A mechanism that allocates scarce developmental work against a composition target defined over sex makes a decision equality law regulates. Directive 2006/54/EC (European Parliament and Council of the European Union, 2006) prohibits direct sex discrimination in access to employment, vocational training, and working conditions, and Directive 2000/78/EC (Council of the European Union, 2000) does the same for religion or belief, disability, age, and sexual orientation; both permit positive action. The

Table S6. Distribution of the $B_k = 4{,}976.64$ protected practice hours in structural inspection, by exact arithmetic. "Composition-blind" distributes protected practice in proportion to the incumbent composition; "Under (S3)" applies the floor with $\theta_{k,g} = 0.32$ and $\varepsilon_k = 0.05$. Certified-equivalents (c.eq.) are hours divided by $h_k = 1{,}440$. The contracted row is zero by construction, because non-payroll practitioners cannot be roster members.

			Composition-blind			Under (S3)		
Group	**Incumbent**	$\theta_{k,g}$	$\Pi_{k,g}$	**h/yr**	**c.eq.**	$\Pi_{k,g}$	**h/yr**	**c.eq.**
Men, permanent	0.875	0.680	0.875	4,354.56	3.024	0.730	3,632.95	2.523
Women, permanent	0.125	0.320	0.125	622.08	0.432	0.270	1,343.69	0.933
Contracted, any gender	—	unset	0.000	0.00	0.000	0.000	0.00	0.000
Total on payroll	1.000	—	1.000	4,976.64	3.456	1.000	4,976.64	3.456

Court of Justice drew the boundary under Directive 76/207/EEC: *Kalanke* (C-450/93) held automatic, unconditional priority unlawful; *Marschall* (C-409/95) upheld a rule with a saving clause requiring individual assessment; *Abrahamsson* (C-407/98) held that preferring a merely sufficiently qualified candidate exceeds the exemption; and *Badeck* (C-158/97) accepted reserved training places where the other group could still train elsewhere. Four design features are chosen against that boundary: the constraint binds an aggregate over a budget period, not an assignment; between rebalances it acts through a dual incentive, not an absolute priority, although a lawful deployment would need an administratively set individual merit floor, encodable as a *restrict* roster guard; it is elastic, so it never compels an assignment; and targets on contract status or career stage, which are not grounds under either directive, raise different questions (fixed-term and part-time work under Directives 1999/70/EC and 97/81/EC, and possible indirect age discrimination for stage-based rules). Whether a given target is proportionate is a fact-specific judgment for the municipality's legal office.

In the Gulf Cooperation Council states, composition targets over nationality are an instrument of labor-market policy. Saudi civil-service appointment is confined to nationals, with non-nationals employed temporarily where no qualified national applicant is available (Kingdom of Saudi Arabia, 1977), and private contractors, through which municipalities procure much sanitation, maintenance, and inspection labor, are graded by the Nitaqat program by the national share of their workforce; Peck (2017) finds that the quota raised national employment at a substantial cost in firm exit. Nitaqat counts posts, so a compliant domain can staff every developmental task with an expatriate lead and form no national successor. With g ranging over nationality, Equation (S3) turns a headcount obligation into a formation obligation. The just-transition ceiling meets the same structure from the other side: the population it cannot see is large, largely migrant, and employed through contractors (International Labour Organization and United Nations Economic and Social Commission for Western Asia, 2026), and extending it beyond the payroll would raise questions under the Personal Data Protection Law (Kingdom of Saudi Arabia, 2021).

S4.3 Data Protection

The capability ledger and the learned beliefs hold per-practitioner practice, recency, proficiency estimates, and, where physiological channels are used, workload responses inferred from biometric signals, retained across multi-year formation periods and used to allocate work. Under the GDPR (European Parliament and Council of the European Union, 2016) this is high-risk processing, and physiological signals may be special-category data, so a data protection impact assessment under Article 35 is mandatory before deployment. The design assumes five measures: *minimization*, since the allocator needs only the readiness test, the due flag, and aggregate shares, so raw physiological signals are reduced on the sensing device to a workload estimate and discarded; *purpose limitation*, so proficiency and workload estimates are used for allocation, training, and safety, never for performance appraisal or discipline; *separation* of protected attributes from the optimizer, which receives only aggregate group shares; *small-cell suppression* through n^{min}; and the *Article 22 safeguards* of the contestability procedure below, whose human re-decision is the required intervention. Participation in physiological monitoring must be voluntary, which is why every multimodal channel in Section 3.4 is optional by construction. These measures align with the privacy-aware learning the Research Topic calls for; federated estimation of beliefs on the practitioner's own device is a natural extension.

S4.4 The Contestability Procedure

Contestability in public-sector algorithms must be a design requirement rather than an audit afterthought (Moon and Guha, 2026; Alfrink et al., 2023). *Standing* extends to affected residents and to any worker whose allocation, developmental access, or displacement is at issue, irrespective of contract status, and to the municipal ombudsman acting on a pattern (Vaccaro et al., 2019). *Windows* are twenty working days to lodge and twenty to resolve, ten with the outcome suspended where a decision is rights-affecting, and forty and thirty for worker appeals on access or displacement, which may surface only when a budget period closes. Every allocation writes an append-only *evidentiary record*: the task profile; the modes and rosters before and after policy filtering with the rules that fired; each candidate's term vector, debt decomposition, and learned success probability with its uncertainty; the stage that selected it; the duals in force; $w_9^\star$; the configuration version; and the accountable human. This is technological due process (Citron, 2008; Citron and Pasquale, 2014) made operational, and recording the configuration makes an appeal against a *policy* possible (Almada, 2019). The *remedy* is re-decision under Zone Z3 by a named human who may depart from the recommendation on the record; an upheld-appeal rate above 0.05 demotes the class one zone and refers it to the review panel. Liability follows the zones: the decision-maker of record in Z3 and Z4, the certifying unit in Z1, and apportionment by the evidentiary record in Z2, since assigning Z2 harms wholly to the vendor would contradict the oversight logic of the EU AI Act (European Parliament and Council of the European Union, 2024).

S5 FURTHER ARITHMETIC OF THE WORKED ALLOCATION

S5.1 Decomposition of the Cost of Preservation

Table S7 decomposes which terms absorb the cost of preservation in the worked domain. The cost figure is a change in a score rescaled against a trailing window, not a share of the inspection budget.

Table S7. Term-by-term decomposition of the cost of preservation: the unconstrained allocation (all tasks to $H{+}A{+}R/a_0$) against the constrained optimum. Cost, energy, and privacy risk are penalties, so a fall is an improvement for those three rows.

Term	Unconstrained	Constrained optimum	Change
Q service quality	0.9100	0.8489	-6.72%
S safety	0.9000	0.8743	-2.85%
P productivity	0.8800	0.7791	-11.47%
Eq^{srv} service equity	0.7400	0.7287	-1.53%
Tr trust	0.6800	0.6713	-1.28%
$Cost$ operating cost	0.5000	0.6226	$+24.53\%$
En energy	0.6000	0.5858	-2.36%
Pr privacy risk	0.3800	0.3574	-5.96%
CAD civic automation debt	0.5004	0.3964	-20.78%
Credited share $\phi_m\psi_a$	0.0000	0.2962	from zero
SCV	0.3426	0.3277	-4.34%

S5.2 Two Parameter Shocks

A retirement wave inside the feasible region. Raising attrition from 0.12 to 0.14, inside the critical rate of 0.1418, expands the budget to $1.2 \times 24 \times 0.14 \times 1440 = 5{,}806.08$ hours and the required credited share from 0.2962 to 0.3456. The same two modes still bracket the requirement, so λ_k stays at 0.0454 and only the mix moves: robot-assisted human work rises from 28.3% to 94.1%, the cost of preservation from 4.34% to 9.57%, and the domain operates at $5806.08/5880 = 98.7\%$ of its ceiling. The arithmetic does not establish that an implementation updating duals at a finite cadence converges to this point.

A retirement wave outside it. At attrition 0.25 the budget becomes 10,368 hours against the ceiling of 5,880, a deficit of 4,488 hours or 3.12 certified-equivalents a year that no allocation of this task stream can form. Two developing practitioners per lead raises $\bar{\zeta}_k$ to $0.70 \times 2/3 = 0.467$ and leaves a deficit of 2,528 hours; $10368/(8 \times 0.35) = 3{,}703$ supervised inspections a year close it; lowering η_k to $5880/(24 \times 0.25 \times$

$1440) = 0.68$ closes it by deciding not to replace the full cadre. The binding quantity is the practice the task stream can carry, not recruitment: with one trainee per inspection the stream supports at most $21{,}000/1{,}500 = 14$ trainee posts at the prevailing automation level.

S5.3 An Analytic Model of Debt Accumulation

Let $p_j(t) = p_{j,\infty}(1 - e^{-t/\tau_j})$ be the fraction of debt component j's required replenishment delivered at time t, and let debt accrue in proportion to the unmet fraction, $\dot{C}_j = c_j[1 - p_j(t)]$. Integrating and weighting by the debt weights gives Equation (S5):

$$C(t) = \sum_j w_j \, c_j \Big[(1 - p_{j,\infty})\, t \; + \; p_{j,\infty}\, \tau_j \left(1 - e^{-t/\tau_j}\right) \Big], \qquad w_j \in \{\alpha, \beta, \gamma, \delta, \epsilon\}. \tag{S5}$$

Table S8 gives illustrative parameters and Figure S1 the trajectories. The model makes one structural point: a capability budget turns *skill* debt from unbounded to bounded, but nothing in the allocation mechanism drives vendor dependency, which keeps accruing linearly. Under these parameters CivicWorkOS's saturating components contribute a bounded 17.98 index units, 11.34 of them skill formation, while its residual accrues at 1.45 units a year, 1.08 of it vendor dependency; Human-First accumulates less debt until year 13.23. The ordering of the curves follows from the replenishment parameters, which encode our reading of each strategy's mechanisms, and is not evidence.

Table S8. Per-component replenishment parameters for Equation (S5), reported as $p_{j,\infty}$ with τ_j in years in parentheses. The accrual rate is $c_j = 10$ index units per year for every component, except that Cost/Performance carries $c_{acct} = 14$ and $c_{dep} = 16$. The final two columns are computed from the row.

Strategy	$p_{skill,\infty}$ (τ)	$p_{fall,\infty}$ (τ)	$p_{acct,\infty}$ (τ)	$p_{dep,\infty}$ (τ)	$p_{trans,\infty}$ (τ)	**Residual slope**	$C(10)$
Automation-First	0	0	0	0	0	10.00	100.00
Cost/Performance	0	0	0	0	0	11.44	114.40
Capability Matching	0.35 (4.05)	0.30 (2.0)	0	0	0	8.36	88.54
Ergonomics-Aware Role Allocation	0.40 (4.05)	0.45 (2.0)	0	0	0	7.89	85.02
Human-First	0.15 (4.05)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	2.38	28.96
CivicWorkOS	**1.00 (4.05)**	**1.00 (1.0)**	**0.85 (1.5)**	**0.10 (2.0)**	**0.95 (1.0)**	**1.45**	**31.51**

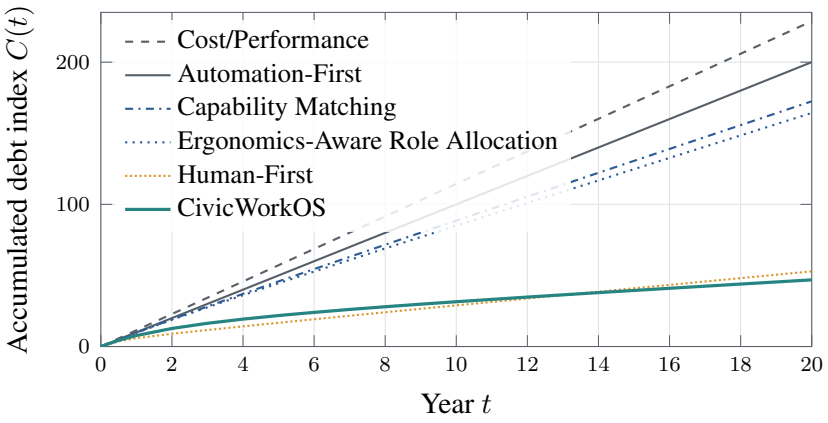


Figure S1. Accumulated Civic Automation Debt under Equation (S5) at the parameters of Table S8. Every point is that equation evaluated at those parameters; nothing is measured or simulated. Human-First and CivicWorkOS cross at $t = 13.23$ years, at 36.73 index units.

S6 ELICITATION PROTOCOL FOR THE JUDGMENT PARAMETERS

Four parameters are structured professional judgments rather than measurements: the learning value $learn_i$, the human developmental share ϕ_m of each hybrid mode, the supervision ratio that bounds ψ_a,

and the certification requirement h_k^{raw}. Two more, the retention time constant τ^{ret} and the residual level ϑ^{res}, can be estimated from the learning layer's own beliefs once a deployment has run, and from the skill-decay literature before it (Arthur et al., 1998; Tatel and Ackerman, 2025). Nothing in this section has been carried out.

Participants. Per domain, at least 12 raters in equal thirds: certified practitioners with at least five years of practice, developing practitioners, and the supervisors who sign off progression. Developing practitioners are the only group with direct experience of what a machine-assisted task teaches.

Instrument. Each parameter is rated on an anchored eleven-point scale. For $learn_i$: 0, no judgment transferable to any other task; 0.5, judgment transferable within the task class only; 1.0, judgment transferable across the domain and required for competence. For ϕ_m: 0, the human observes an outcome and cannot alter it; 0.5, the human sets the parameters of a machine determination and reviews its output against independent evidence; 1.0, the human forms the determination and the machine supplies data only. For the supervision ratio, the maximum number of developing practitioners one lead can supervise without compromising their own attention, conditioned on risk and criticality. For h_k^{raw}, the documentary hours required and whether they are logged as unshared, unassisted practice, and separately the hours actually needed.

Procedure and reliability. Two Delphi rounds, the second showing the anonymized first-round distribution and rationales. Agreement is assessed by ICC(2,k) against an acceptance threshold of 0.75; a parameter below it is reported as a range and the sensitivity analysis is rerun over the observed range.

S6.1 Ethics and Data Protection

The protocol involves human participants judging their own work and their colleagues' development, so it requires institutional ethics approval and written informed consent. Ratings are held pseudonymously, only group-level results are reported, and no rating is disclosed to an employer in identifiable form. The consent form must state that refusal has no effect on the participant's allocation under any deployed system.

S7 EXTENDED RELATED WORK

S7.1 Human–Robot and Human–AI Task Allocation

Ranz et al. (2017) matched task requirements to human and robot capabilities, and Dorri et al. (2018) survey the multi-agent coordination mechanisms that allocation markets build on. El Makrini et al. (2019) considered ergonomics in collaborative assembly, and Merlo et al. (2023) allocate roles dynamically over AND/OR graphs with ergonomic risk estimation. Noormohammadi-Asl et al. (2025) incorporate leading and following preferences, Ali et al. (2022) re-optimize allocation as modeled trust evolves, and Chen et al. (2018) plan with a learned trust model in a partially observable decision process. Gombolay et al. (2015) show that the allocation of decision authority between human and robot affects both team efficiency and worker satisfaction. Ajoudani et al. (2018) review human–robot collaboration and Lasota et al. (2017) its safety, Meseguer Valenzuela and Blanes Noguera (2025) review allocation for mobile robot fleets, Hady et al. (2025) multi-agent reinforcement learning for resource allocation, and Zhao et al. (2023) allocate spatio-temporal sensing tasks across a mobile crowd at city scale. Personnel rostering carries skill mix and qualification maintenance as hard constraints (Burke et al., 2004; Van den Bergh et al., 2013), the discipline Section 3.6 adopts; it differs in assuming that the human-versus-machine decision has already been made.

S7.2 Adaptive Automation, Neuroergonomics, and Trust

Sheridan and Verplank (1978) supplied the levels-of-automation scale, generalized by Parasuraman et al. (2000) across information acquisition, analysis, decision, and action, and Shneiderman (2020) recasts automation and human control as independent axes. Endsley and Kiris (1995) formalized the out-of-the-loop performance problem, Parasuraman and Riley (1997) organized automation failures into use, misuse, disuse, and abuse, and Kaber and Endsley (2004) showed that intermediate and adaptive automation preserve situation awareness. Parasuraman (2003) defined neuroergonomics as the study of the brain at work, Dehais et al. (2020) review neurophysiological markers of workload and engagement,

Debie et al. (2021) review multimodal fusion for workload assessment, and Aricò et al. (2016) close the loop with an EEG-triggered adaptive automation in air-traffic control. NASA-TLX remains the subjective reference (Hart and Staveland, 1988). Lee and See (2004) define appropriate reliance, and Wischnewski et al. (2023) show that miscalibrated trust degrades team performance independently of system accuracy.

S7.3 Skill Decay, Learner Models, and the Ironies of Automation

Bainbridge (1983) identified the ironies of automation, Arthur et al. (1998) meta-analyzed skill decay over non-use intervals, and Tatel and Ackerman (2025) update the estimate for procedural skills. Jaber and Bonney (1996) model learning interrupted by production breaks. Corbett and Anderson (1994) introduced Bayesian knowledge tracing and Pelánek (2017) review its logistic successors, of which the outcome model of Section 3.4 is a team-level relative; Glickman (1999) and Minka (2001) supply the online Bayesian updates. Zuboff (1988) distinguishes automating from informating work, and Rashidi (2026) argues that AI can automate the entry-level work through which judgment forms.

S7.4 Control Transfer and Competence Awareness

Gold et al. (2013) measured takeover time against time budget, Zhang et al. (2019) meta-analyzed its determinants across 129 studies, and SAE J3016 (SAE International, 2021) defines the minimal-risk condition. Basich et al. (2020) let an autonomous system learn its competence at each level of autonomy and seek human help where needed, and Norton et al. (2022) propose metrics by which robots assess and communicate their proficiency. The ISO 10218 series sets safety requirements for industrial robots and their integration (International Organization for Standardization, 2025a,b).

S7.5 Municipal Governance, Labor, and Contestability

The task-based framework (Autor et al., 2003) underpins the argument that displacement is overstated when complementarities are ignored (Autor, 2015); occupation-level exposure estimates (Frey and Osborne, 2017) fall roughly fivefold at task level (Arntz et al., 2016); language models widen exposure (Eloundou et al., 2024); and robots lower employment and wages for lower-skilled workers (Acemoglu and Restrepo, 2020). Platform workers bear exposure outside municipal protections (Wood et al., 2019; De Stefano, 2016), automated public-service decisions concentrate errors on the poorest (Eubanks, 2018), and public services are constituted by the discretion of street-level bureaucrats (Lipsky, 2010). Algorithmic management (Milanez et al., 2025; OECD, 2026), ethical frameworks for human–robot collaboration (Callari et al., 2024), the ILO just-transition guidelines (International Labour Organization, 2015), and regional projections (International Labour Organization and United Nations Economic and Social Commission for Western Asia, 2026; World Economic Forum, 2025) frame the institutional response. Contestability draws on technological due process (Citron, 2008; Citron and Pasquale, 2014), the right to meaningful human intervention (Almada, 2019; Vaccaro et al., 2019), and design frameworks for contestable AI (Alfrink et al., 2023). City digital twins assist policy decisions (Temple et al., 2024), and Diaz-Sarachaga (2025) proposes criteria for whether a deployed twin is well governed. Training economics explains why employers under-provide portable on-the-job training (Acemoglu and Pischke, 1998; Wolter and Ryan, 2011), and Sen (1999), Nussbaum (2011), and Ostrom (1990) supply the vocabulary of capability and of monitored appropriation rules over depletable shared resources.

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